

① Let $x = 2^3 + 2^{-9} + 2^{-22}$ find machine number on the
 more-32 that right on left x . More number

$$x = (1, a_1 \dots a_{23}) \times 2^m \text{ where } -126 \leq m \leq 127$$

$$x = q \times 2^m \quad 1 \leq q < 2 \quad -126 \leq m \leq 127$$

$$(1, a_1 \dots a_{23})_2 \times 2^m$$

$$x_{\text{left}} = (1 \dots a_{23})_2 \times 2^m$$

$$x_{\text{right}} = ((1 \dots a_{23})_2 + 2^{-23}) \times 2^m$$

$$x_{\text{left}} \leq x \leq x_{\text{right}}$$

Answer

$$\begin{array}{r} 0.000000000000000000000001 \\ + 0.000000000000000000000001 \\ \hline x = 1,00,0.000000000000000000000001001 \\ 1, \underset{1}{0} \underset{2}{0} \underset{3}{0} \underset{4}{0} \underset{5}{0} \underset{6}{0} \underset{7}{0} \underset{8}{0} \underset{9}{0} \underset{10}{0} \underset{11}{0} \underset{12}{0} \underset{13}{0} \underset{14}{0} \underset{15}{0} \underset{16}{0} \underset{17}{0} \underset{18}{0} \underset{19}{0} \underset{20}{0} \underset{21}{0} \underset{22}{0} \underset{23}{0} \times 10^3 \end{array}$$

$$m = 3$$

$$x_{\text{left}} = (1.000000000000000000000010) \times 2^3$$

$$x_{\text{right}} = (\dots \dots \dots) + 2^{-23}) \times 2^3$$

SO2U 2

find way to compute the following function without series loss of significance.

$\log(x) - \log(y)$ Assume x and y are positive number and x is close y .

$$\log(x/y) = \log(x) - \log(y)$$

$$\log(x/y) = \log x(x) - \log(y)$$

$$\log(x/(x+y)/2) + \log(((x+y)/2)/y).$$

x close y

$$\log(x/y) = \log(x/y) - \log(y/(x+y))$$

$$\log(x/y) - \log(y/(x+y))$$

SO2U 3

$$\text{lex } x = 0.642676895$$

$$y = 0.642661224$$

8ix digit

mantissa, relative err.

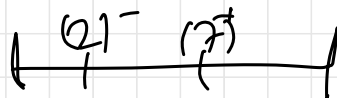
$$x - y = 0.00001566179$$

$$0.000016 \quad) b$$

$$\text{Absolute} = |a - b|$$

$$\text{relative err} = \text{absolute} / a \times 100 = \%2.161$$

4) $f(x) = x^3 - 5x^2 - x + 5$ Bisection method $a_0 = 2$
and $b_0 = 7$ calculate?



$$f(2) = 8 - 20 - 2 + 5 = -9 \quad (-)$$

$$f(7) = 243 - 245 - 7 + 5 = 96 \quad (+)$$

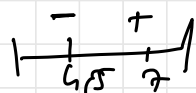
$$\boxed{f(2) \cdot f(7) < 0} \quad - \text{ aralarında kök var.}$$

adım 2

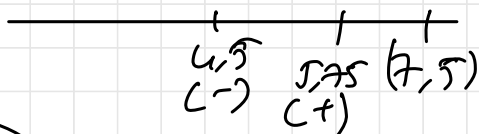
$$\frac{2+7}{2} = 4,5 \quad f(4,5) = 91,125 - 101,25 - 4,5 + 5 = -9,625$$

adım 3

$$\frac{4,5 + 7}{2} = 5,75$$



$$f(5,75) = 190,109 - 165,212 - 5,75 + 5 = 24,047$$



adım 4

$$\frac{4,5 + 5,75}{2} = 5,125$$

$$f(5,125) = 134,61 - 131,228 - 5,125 + 5 = 3,15$$

adım 5

$$\frac{4,5 + 5,125}{2} = 4,8$$

$$101,59 - 115,2 - 4,8 + 5 = -9,4$$

adm 6

$$\frac{5,12 + 4,8}{2}$$

5

18K

$$125 - 125 - 5 = \emptyset$$

SOLU 5

Newton Raphson method

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x) = x^3 - 5x^2 - x + 5$$

$$f'(x) = 3x^2 - 10x - 1$$

$$x_0 = 7$$

2. adm

$$n=0 \text{ i.e. } x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\Rightarrow x_1 = 7 - \frac{f(7)}{f'(7)}$$

$$\Rightarrow \frac{242 - 245 - 7 + 5}{147 - 70 - 1} = \frac{96}{76}$$

$$x_1 = 5.73$$

Galima Notlari $\rightarrow$ single, 8bit, 24bit

Nearly machine Number

$$x = q * 2^m \quad 1 \leq q < 2 \quad -126 \leq m \leq 127$$

what is the number machine

$$x = (1, a_1, a_2, a_3 \dots a_{23})_2 * 2^m$$

$$x = (1, a_1 \dots a_{23})_2 + 2^{-23} * 2^m$$

absolute error
 $real = x - y$

$$relative = \frac{real - fl(x - fl(y))}{real}$$

Taylor

$f(x) = e^x$ in $x=1$ taylor series

$$f(1) = e \quad f'(x) = e^x$$

$$f'(1) \cdot (x-1) = e \cdot (x-1)$$

$$f''(1) \cdot \frac{(x-1)^2}{2!} = e \cdot \frac{(x-1)^2}{2!}$$

Bisection

$$\frac{a+b}{2} = \text{yerik key}$$

$$f(a) f(b) < 0$$

$$f(a) f(b) = \text{yerik key}$$

Newton method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad n \geq 0$$



$$x = \sqrt{2} \Rightarrow x^2 - 2 = 0$$

$$f(x) = x^2 - 2$$

$$f'(x) = 2x$$

gerade kkk

$$|p - x_n| < \text{TOL}$$

gerade kkk

$$|x_{n+1} - x_n| < \text{TOL}$$

$$\frac{|x_{n+1} - x_n|}{|x_{n+1}|} < \text{TOL}$$

Qr2k

Example

$$52.21875 \Rightarrow \underline{110100.00111}$$

$$= 11010000111 \times 2^{-6} \quad \text{m}$$

23bit mantissa

$$\begin{array}{c} \text{sing} \quad 0.110100001110000000000000 \\ \hline \text{mantissa} \end{array}$$

$$127 + m \Rightarrow 6 \Rightarrow \text{Exponent} \Rightarrow 133$$

$$52.21875 \Rightarrow 0.11010000111 \times 2^{133} \Rightarrow 10000101$$

$$\begin{array}{c} \text{32bit sing used} \quad 0.1000101 \\ \hline \text{sig} \quad \text{exp} \quad \text{mantissa} \end{array}$$

isoleksn pozitif

Example

$$52.21875 \Rightarrow 110100.00111 \Rightarrow$$
$$1.1010000111 \times 2^5 \Rightarrow$$

$$\text{Exponent} = (e - 126) = 5 \quad e = 132$$

